CASE B · THE DATA MOAT

How the Financial Model Was Prepared

Method, evidence, tests and audit trail behind the model,
for judges, mentors and teammates who want to see the work.

Team AlphaQ · Nextute Edtech × Finance Club, IIT Patna · Case Competition 2026

What this report is

The record of how the Data Moat financial model was prepared: what it had to do, how the work was done, what evidence stands behind each input, how the calculations work, how they were tested, what an independent CFO review changed, and how to reproduce everything. It complements the plain-English guide, which explains what each tab means.

The work in numbers

102 inputs, each with unit, range, source and confidence	**7** research workstreams (R1 to R7)	**18** findings fixed in the CFO audit
90 values matched by an independent rebuild	**23** automatic error checks, all passing	**0** formula errors in about 37,000 cells

CONTENTS

About this report

IN PLAIN ENGLISH

In one paragraph. We built a live-formula Excel model for an AI analytics product sold to coaching institutes. It answers three questions: is the product worth its price to an owner, do we earn our money back on each institute, and what does that add up to over three years. We then had it reviewed like a sceptical CFO, found 18 problems, fixed them, and re-tested. The honest answer: on hard benefits an owner gets back **0.55x** of the cost (1.7x if admissions improve); each institute is worth **3.6x** what it costs to win; the business is lean and market-capped (₹65.3 lakh revenue and ₹45.0 lakh of funding over three years).

FROM THE PS (the case brief)

Why this matters for the competition. PS Section 10 gives 20% of the marks to the revenue model and unit economics (including the 3-year projection) and says teams that state assumptions and limitations clearly score higher than teams that oversell. This report is the evidence that the numbers are traceable, tested and honest.

Chapters

Part 1 · What the model had to do 3

Part 2 · How it was prepared 4

Part 3 · How the model is built 5

Part 4 · The evidence behind the inputs 6

Part 5 · The method, module by module 7

Part 6 · How it was tested 8

Part 7 · The CFO audit 9

Part 8 · What the model says 10

Part 9 · Limits, open items and how to reproduce 11

PART 1

What the model had to do

Everything starts from the problem statement (the PS). We traced each requirement to a place in the workbook before building anything, and re-traced it during the audit.

PS requirement	PS ref	Where it is answered	Status
Challenge the naive worked example (2.25x return, 10x LTV/CAC)	Sec 6	A "Guide (naive)" column beside our numbers on every output tab	Done
Measurable ROI for an institute; who is the buyer	Sec 3 · B1	Institute_ROI: three returns and a break-even	Done
Pricing, packaging, CAC, payback, churn, LTV	Sec 3 · B2	Pricing, COGS_Build, Unit_Econ	Done
Cities and sizes first; direct, partners or product-led	Sec 3 · B3	Unit_Econ by size and channel; Projection market cap	Done
Infrastructure + AI cost per student per month	Sec 3 · A7, Sec 7	COGS_Build: ₹3.62 against the PS range ₹3–10	Done
Churn after exam season; seasonal student counts	Sec 6	Seasonality and the renewal logic in Unit_Econ	Done
Live formulas, listed assumptions, at least one sensitivity	Sec 9 · Round 1	All tabs; Assumptions and Sources; Sensitivity	Done
3-year projection, scenarios, sensitivities	Sec 9 · Round 2; Sec 4 and 10	Projection, Scenarios, Sensitivity	Done; 3-year sensitivity grid still to come
Ranges, scenarios and evidence	Sec 6	Bear / Base / Bull on 13 drivers, a Stress test, a source and confidence per input	Done
State assumptions and limitations	Sec 10	Warnings on Checks; evidence labels; the limits in chapter 9	Done
Other revenue models (outcome-linked, parent, marketplace, benchmarks)	Sec 6	Not modelled; listed as such	Open by choice

WATCH OUT

Out of scope on purpose: upsell and expansion revenue (so no NRR), income tax, early cash from annual prepay, any management response in the Bear case, and the PS revenue models other than per-student SaaS.

PART 2

How it was prepared

Eight steps, in order. Each step left a file behind, so a reviewer can check the work.

#	Step	What was done	What it left behind
1	Challenge the guide (20 Oct)	Listed the ten weakest points of the PS's naive example: flat 25% prevention, whole-fee loss, 12 billing months, 1/churn lifetime, one average institute, no seasonality, no proof.	FINANCE_PLAN.md
2	Research (20 and 22 Oct)	Seven workstreams, run in parallel: R1 competitor prices, R2 fees and refund rules, R3 dropout evidence, R4 trials on early warning, R5 SaaS benchmarks and salaries, R6 unit costs (WhatsApp, AI, GST, payments, cloud), R7 market size. Rules: primary sources first, never estimate (write “not found”), every row with a link, date and confidence.	research/R1 to R7 (.md)
3	Design the inputs	Every number became a row with unit, Bear / Base / Bull, an evidence type (Researched, Public, Guide, Assumption, Structural), a confidence (H, M, L), a source and a note.	assumptions.csv, seasonality.csv
4	Build (20 Oct)	A Python generator writes the workbook: live formulas only, one Assumptions sheet, named ranges, a scenario switch, a hidden Engine that re-runs the model for scenarios and sensitivities.	build_model.py, the .xlsx
5	First verification	Automatic checks in the workbook; an independent plain-Python rebuild of the unit economics matched to the rupee.	Checks tab, verify_model.py
6	Document	Slide-ready numbers with cell references, a “do not claim” list, and a plain-English guide to every tab.	README.md, the explainer PDF
7	CFO audit (22 Oct)	Re-read the PS, cross-checked every input against the technical team's measurements and against how SaaS peers calculate. Found 18 issues, including a missing 3-year projection. Fixed them, added the Projection tab and the market research.	MODEL_AUDIT.md, v1.1
8	Re-verify and hand off	Full recalculation, independent rebuild of ROI, unit economics and the projection, a structural scan, refreshed documents.	this report, DECK_NUMBER_UPDATES.md

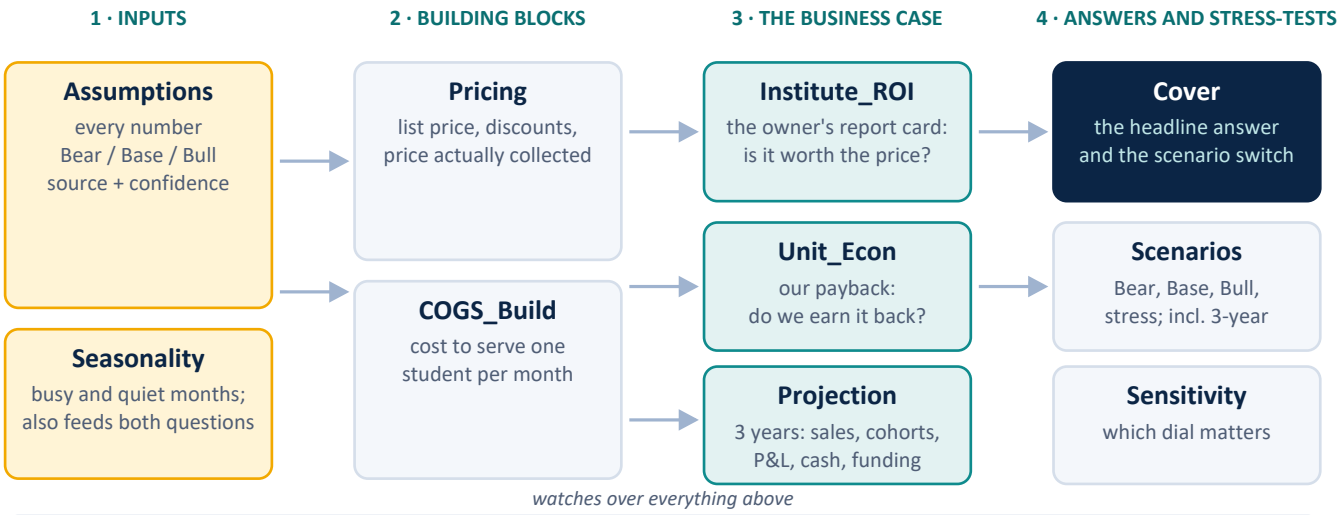
IN PLAIN ENGLISH

The rule behind every step: a number is either sourced, or labelled as an assumption, or both. Nothing is typed into a calculation; if it changes, it changes in one place and everything follows.

PART 3

How the model is built

A small set of design rules does most of the work of keeping a 30,000-formula workbook trustworthy.



Inputs flow right: Assumptions and Seasonality, then Pricing and COGS, then the business case, then the answers. Checks and Sources watch over everything.

Design rule	Why it matters
One home for numbers: the Assumptions sheet (plus the monthly calendar on Seasonality)	A judge can change one cell and see everything move; nothing is hidden inside a formula
The workbook is generated from two CSV files by a script	Inputs are versioned and diffable; the script rebuilds the workbook in seconds
Colour code: blue = typed input, black = formula, green = pulled from another tab	A reader can tell at a glance what is an assumption
Scenario drivers: only 13 inputs move with the Bear / Base / Bull switch	Scenarios are coherent stories, not an everything-at-worst joint extreme
A hidden Engine and three hidden Proj_ sheets hold copies of the whole chain	Scenarios, Stress and the tornado work without macros or Excel Data Tables, which break in Google Sheets
Each calculation sheet starts with the inputs it uses, pulled in green	No typed number ever sits inside a calculation
Checks sheet with error checks and honest warnings	Arithmetic errors stop the Cover from saying ALL CHECKS PASS; uncomfortable results raise a warning
Row positions of every output are exported to model_layout.json	Downstream documents read exact cells instead of guessing at labels

PART 4

The evidence behind the inputs

Of 102 inputs, 23 are researched, 1 are public, 16 come straight from the PS, **57 are our own assumptions** and 5 are modelling conventions. Half the model is still judgement, and it says so.

Workstream	What it established	What it fed in the model
R1 · Competitor prices	ERP and LMS tools list at about ₹4–21 per student per month at 400 students (Classplus, Fedena, Graphy); nobody bundles OMR, topic analytics and Hindi parent reports	Price anchor; the warning that our ₹30 list price is above every competitor
R2 · Fees and refund rules	Local institutes charge about ₹50k–1.2L a year; the Ministry of Education guideline (cl. 8(iv)) and Rajasthan Act (s.6(2)(d), s.13(4)) set instalment and pro-rata refund rules, but chains rarely follow them	Fee levels by size (₹60k / ₹75k / ₹100k), instalments, refund compliance
R3 · Dropout evidence	Thin: a trade body says “at least 10% per batch”; no timing or reasons by share. The PS baseline is 8–15%	Dropout rate 12%; timing curve flagged as an assumption
R4 · Trials on early warning	Eight randomised trials: mean 15.8%, median 12.5% relative reduction in dropout among flagged students who get outreach; India's Room to Read 30%, MAAPS and Samata nulls	Trial effect 15% (5% / 15% / 30%), labelled “borrowed from other settings”
R5 · SaaS benchmarks, salaries	SaaS Capital 2025 net revenue retention 90 / 98 / 106%; ChartMogul churn; India SMB CAC and payback only from weak sources; Tier-2 salaries from Payscale and job postings	Churn, conversion, CAC build, salaries; most rows flagged confidence L
R6 · Unit costs	WhatsApp utility ₹0.115–0.145 a message; GST 18% (unregistered institutes cannot reclaim); hosted AI prices; Razorpay 2.36%; cloud prices	COGS ₹3.62 per student per month; GST and gateway inputs
R7 · Market size	No institute counts by size for any state; 574 registered coaching institutes in Patna district; JEE and NEET registrations; SAM derived bottom-up	Addressable market 1,180 and penetration 8%, both assumptions
Technical team's own measurements	Recall 0.31–0.41 of dropouts at a workable workload; precision about 0.15; 78% of students are minors; dropout peaks in months 3–4; 25% of dropouts are sudden	Recall 32% / 42%, precision 15%, minors share, timing

WATCH OUT

Where the evidence is weakest: willingness to pay, the admissions uplift, pilot-to-paid conversion, the addressable market, mentor time per flag, and the upfront-payer and refund-compliance shares. These are labelled Assumption, shown in the tornado, and listed as the first things to test in owner interviews.

PART 5

The method, module by module

What each calculation block does, with the Base-case numbers for a typical 400-student institute.

5.1 Owner's return (Institute_ROI)

- **What a dropout costs.** For each month a student might leave, the fee actually lost is the unpaid instalments (instalment payers) or the pro-rata refund times its compliance rate (upfront payers); weighted by the dropout-timing curve it averages ₹29,799, not the whole ₹75,000 fee.
- **The prevention funnel.** Share prevented = share catchable after the blind months (87%) × recall (34.2%) × follow-up rate (90%) × trial effect (15%) = **4.0%**, not the PS's flat 25%.
- **Value and cost.** Value = fees retained (₹53,903) + staff time (₹30,000) + admissions margin (₹1,80,000). Cost = platform fee (₹1,06,546) + GST not reclaimed (₹7,671) + mentor time acting on flags (₹39,398, 27% of the roll flagged).
- **Three returns.** Fees only 0.35x; hard benefits (fees + staff time) 0.55x; all benefits 1.7x. Break-even needs 5.5 students kept or 1.5 extra admissions a year.

5.2 Our unit economics (Unit_Econ)

- **Revenue and contribution.** Price × active students × a monthly seasonal index, with a minimum fee, discounts and bad debt: ₹8,612 a month; after cost of serving and customer-success staff, ₹6,740 (78%).
- **Acquisition.** Rep cost with 30% overhead ÷ pilots per rep, ÷ pilot-to-paid conversion (30%), plus marketing: CAC ₹15,567. Adding pilot cost and onboarding: ₹38,392 fully loaded.
- **Retention.** Monthly churn applies only to institutes not on annual prepay (1.8% a month); a renewal spike (20%) applies in three renewal months, and only after nine months of paying. Annual retention 64%.
- **LTV and payback.** Contribution × share still paying × discount factor over 60 months: ₹1.40 lakh discounted, ₹1.85 lakh undiscounted. LTV ÷ CAC 3.6x and 4.8x; payback 5.7 months simple, 6 survival-weighted. A channel partner lifts small institutes from 1.5x to 3.4x.

5.3 The 3-year projection (Projection)

- **Sales capacity.** Reps × pilots per rep × a sales-intensity index that peaks March to June; pilots convert a month later at 30%.
- **Market cap.** New institutes never exceed addressable institutes × maximum penetration = 94 in total; the cap limited sales in 6 months.
- **Cohorts.** Each month's new institutes are tracked separately with the same churn and renewal logic as Unit_Econ.
- **Costs.** Variable cost of serving, server stacks, customer-success staff in steps, sales and marketing, engineers, founders and overheads, with percentage-of-revenue floors so costs grow with revenue.
- **Cash.** Receivables, equipment for new hires, then the funding ask = deepest cash deficit (₹27.5 lakh) + three months of costs (₹13.1 lakh), rounded up: **₹45.0 lakh**.

5.4 Scenarios and sensitivity

- **Scenarios.** Bear / Base / Bull flex the 13 drivers marked Y on Assumptions; all else stays at Base. **Stress** flexes every input to its worst and is a survival test.
- **Tornado.** 17 inputs, each moved alone from its Bear to its Bull value; the chart ranks them by swing in the owner's return and in LTV ÷ CAC.
- **Two what-if grids.** The owner's return by price and prevention rate; LTV ÷ CAC by churn and acquisition cost. Both are computed with plain formulas, not Excel Data Tables.

PART 6

How it was tested

A green recalculation proves the formulas evaluate, not that they are right. So there are four independent layers.

Test	What it proves	Result
Full recalculation of the workbook in a separate engine (no Excel)	No broken references, no #errors, no stray artifacts	About 37,000 cells, 0 formula errors
23 automatic error checks inside the workbook	Percentages in range, timing sums to 100%, customer mix sums to 100%, live values inside Bear–Bull, projection never exceeds the market cap or runs out of cash	All pass
Engine reproduces the visible tabs for all three sizes (6 checks)	Scenarios and the tornado use exactly the same logic as the main tabs	All pass
Our copy of the PS example reproduces 2.25x	We read the case brief correctly	Pass
Independent plain-Python rebuild of ROI and unit economics (3 sizes)	The spreadsheet formulas compute what the method says	57 of 57 values match
Independent rebuild of the 3-year projection (Bear, Base, Bull)	Cohorts, market cap, P&L, cash and funding ask are right	33 of 33 values match
Live Projection equals the fixed-scenario sheet for the chosen scenario	The hidden copies agree with the visible one	Pass
Structural scan of 31,232 formulas and 117 named ranges	No leftover placeholders; no names that Excel could confuse with cell references	Clean
Cross-check with the technical team's measurements and with peer conventions	Inputs and definitions agree across workstreams	Led to findings F3 to F11

FROM THE PS (the case brief)

The three warnings that fire today are limitations the model reports about itself, which PS Section 10 rewards: list price above the highest erp/lms anchor found (₹21/student/month); mid hard-benefit roi (fees + staff time) below 1.0x; the market cap limited sales in the live projection.

WATCH OUT

What testing cannot do: prove the assumptions true. The checks confirm the arithmetic and the logic; only owner interviews, a pilot and the technical team's measured metrics can confirm the inputs.

PART 7

The CFO audit

On 22 October the model was re-read as a sceptical CFO would read it. The full record is in MODEL_AUDIT.md; this is the summary.

#	Finding	Fix
F1	A May sign-up faced the renewal spike in June (LTV understated about 7%)	Renewal only after 9 months of paying
F2	PS column used our fee instead of the PS's ₹60k for "cost as % of fee"	Added the PS's own fee input
F3	Mentor time 30 minutes per flagged student	120 minutes
F4	Recall 55% / 65% versus 31–41% measured by the tech team	32% / 42%
F5	Precision 35% versus about 15% implied by the tech data	15%; alert-load check added
F6	Dropout rate used a trade-body floor as a mean	PS baseline 12%
F7	Dropout timing front-loaded; PS and tech data peak in months 3–4	Curve re-shaped; two blind months
F8	Minors share 75% versus 78%	78%
F9	Mid fee came from chain price pages	₹60k / ₹75k / ₹100k
F10	No simple payback or undiscounted LTV (what peers and the PS quote)	Both added
F11	CAC excluded sales overhead and sat below the PS's ₹15–30k range	30% loading
F12	"Gross margin" excluded support staff	Relabelled; contribution margin = peer gross margin
F13	An unproven assumption carried two-thirds of the headline return	Hard-benefit return leads
F14	Bear and Bull moved all inputs at once	13 drivers plus a Stress column
F15	No 3-year projection (the PS rubric lists one)	New Projection tab
F16	No channel-partner economics (PS question B3)	Field versus channel comparison
F17	Draft projection let costs stay flat as revenue grew	R&D; and overhead floors
F18	No market cap on the projection	Cap from research R7; one-rep hiring plan

What moved (Base, 400-student institute)

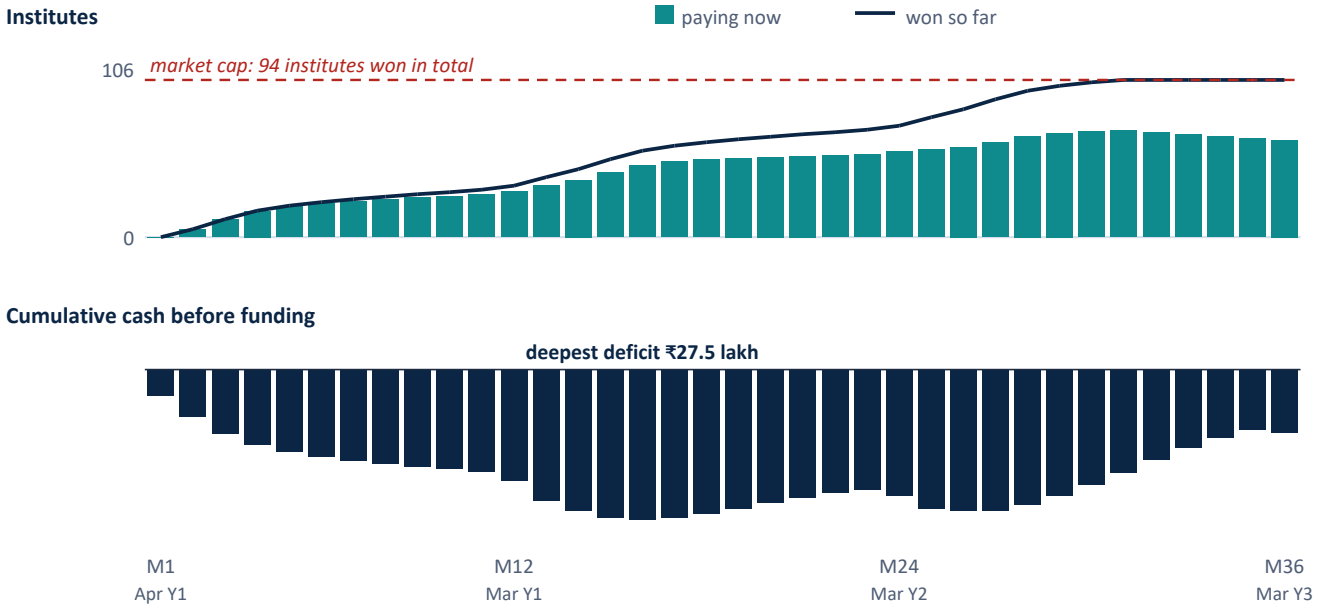
Metric	v1.0	v1.1
Share of dropouts prevented	6.6%	4.0%
Owner's return: fees only / hard benefits / all benefits	0.75x / about 1.0x / 2.9x	0.35x / 0.55x / 1.7x
Students to keep to break even	3.5	5.5
CAC (sales + marketing)	₹12,667	₹15,567
LTV ÷ CAC, discounted	3.7x	3.6x
Year-3 revenue / EBITDA / funding	not modelled	₹65.3 lakh / ₹11.9 lakh / ₹45.0 lakh

PART 8

What the model says

The corrected model is less flattering and more defensible. Bear, Base and Bull flex the key drivers; Stress is a survival test.

Mid institute	Bear	Base	Bull	Stress
Owner's return, all benefits	0.37x	1.7x	3.9x	0.09x
Owner's return, hard benefits	0.37x	0.55x	1.3x	0.09x
Owner's return, fees only	0.09x	0.35x	1.1x	0.02x
LTV ÷ acquisition cost, discounted	0.75x	3.6x	10.5x	-0.11x
LTV ÷ acquisition cost, undiscounted	0.93x	4.8x	14.8x	-0.13x
Fully loaded acquisition cost	₹82,540	₹38,392	₹25,281	₹1,90,892
Revenue in year 3	₹11.0 lakh	₹65.3 lakh	₹2.50 crore	n/a
EBITDA in year 3	-₹30.2 lakh	₹11.9 lakh	₹1.26 crore	n/a
Funding the model says we need	₹1.00 crore	₹45.0 lakh	₹30.0 lakh	n/a
Break-even month	after month 36	month 16	month 4	n/a



Base case: paying institutes (bars), institutes won so far (line), the market cap (dashed); cumulative cash before funding below.

- **Retention alone does not pay for the product.** Fees retained return 0.35x of the owner's cost. The pitch must be a bundle: staff time, parent reports, results.
- **Our unit economics hold.** LTV ÷ CAC 3.6x discounted, 4.8x undiscounted; simple payback 5.7 months. Small institutes belong with channel partners (3.4x against 1.5x by field sales).
- **The market caps the plan, not money.** At 8% of 1,180 institutes the business is lean; a second sales rep does not help until territory grows.
- **Price is a two-sided trap.** A lower price helps the owner's return and hurts our LTV ÷ CAC. Only owner interviews can find the price where both work.

PART 9

Limits, open items and how to reproduce

What a reader should not over-trust, what to do next, and the commands that rebuild everything.

Limit	Why it matters	How it closes
Willingness to pay (₹30 list)	Above every competitor price found; untested	Owner interviews and a Van Westendorp price survey
Admissions uplift (1%)	Lifts the owner's return from 0.55x to 1.7x; no evidence	Track next-session admissions in pilots
Addressable market and penetration	Derived, not counted; caps the whole 3-year plan	10–15 owner calls; the UP district survey when published
Recall and precision	Placeholders matching the tech team's early tests	Replace with Agent C's metrics.json
Pilot conversion, mentor time, support load	Borrowed or assumed	Time-track the first 20 pilots
Bear funding	Assumes no cost response	A real team would cut costs
Not modelled	Expansion revenue and NRR, outcome-linked pricing, tax, prepaid cash timing, a 3-year sensitivity grid	Round 2 extras

Reproduce everything

Command (in the Financial_Model folder)	What it does
python build_model.py	Rebuilds DataMoat_Financial_Model_v1.1.xlsx from assumptions.csv and seasonality.csv
python verify_model.py --all	Recalculates without Excel; prints the Cover, the Checks, Scenarios, Sensitivity and the projection (about 3 minutes)
python guide/extract_numbers.py	Reads every output cell into guide_numbers.json (about 4 minutes)
python guide/build_guide_pdf.py	Rebuilds the plain-English explainer
python guide/build_prep_report.py	Rebuilds this report

IN PLAIN ENGLISH

Sign-off. Model v1.1 CFO audit · 2026-10-22. Verified: 0 formula errors, 23 of 23 error checks pass, 90 of 90 values reproduced independently. Records: MODEL_AUDIT.md (the audit), README.md (slide numbers), DECK_NUMBER_UPDATES.md (old to new numbers), research/ (sources). Every figure here is a model output, not a measurement.